OFFICIAL TECHNICAL SPECIFICATION DATASHEET
MANUFACTURER: REC GROUP
MODEL / SERIES: REC Alpha Pure-R (REC420AA Pure-R)
DOCUMENT TITLE: REC Alpha Pure-R Series Heterojunction Module Technical Datasheet
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1. PRODUCT IDENTIFICATION & ARCHITECTURE
  Cell Technology: Heterojunction (HJT)
  Module Construction: Monofacial
  Number of Cells / Format:80 half-cut REC heterojunction cells with lead-free gapless techno
  Glass / Front Encapsulant:3.2 mm solar glass with anti-reflective surface treatment
  Frame Material: Anodized aluminium (black) with cross-support bars
  Junction Box / Protection: IP68 with 4 bypass diodes

2. ELECTRICAL CHARACTERISTICS AT STC (1000 W/m2, 25C, AM 1.5)
  Nominal Maximum Power (Pmax / Pnom): 420.0 Wp
  Module Aperture Efficiency: 21.80 %
  Open Circuit Voltage (Voc): 59.40 V
  Short Circuit Current (Isc): 8.88 A
  Maximum Power Voltage (Vmp): 50.00 V
  Maximum Power Current (Imp): 8.40 A
  Maximum System Voltage: 1000.0 V DC
  Maximum Series Fuse Rating: 25.0 A

3. TEMPERATURE CHARACTERISTICS & OPERATING LIMITS
  Temp. Coefficient of Pmax (gamma): -0.240 %/C
  Temp. Coefficient of Voc (beta): -0.240 %/C
  Temp. Coefficient of Isc (alpha): +0.040 %/C
  Nominal Cell Temperature (NOCT/NMOT): 44 +/- 2 C (NMOT)
  Operating Temperature Range: -40 C to +85 C

4. MECHANICAL DIMENSIONS & RATINGS
  Length: 1730.0 mm
  Width: 1118.0 mm
  Thickness (Height): 30.0 mm
  Gross Aperture Area: 1.9341 m2
  Module Weight: 21.5 kg
  Mechanical Design Load (Front): 5400 Pa (Snow load)
  Mechanical Design Load (Rear): 2400 Pa (Wind load)

5. WARRANTY & LIFETIME PERFORMANCE
  Product Workmanship Warranty: 20.0 Years
  Linear Power Output Warranty: 25.0 Years
  Annual Linear Degradation: 0.25 % / year
  End of Warranty Retained Output: 92.0 %

6. PROVENANCE & ARCHIVAL METADATA
  Authoritative URL: https://www.recgroup.com/sites/default/files/documents/ds_rec_alpha_pure
  Document Verification: OFFICIAL_FAMILY_DOCUMENT
  Installation Mapping: BENCHMARK_REFERENCE_MODULE
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SolarSelect MLOps Hardware Database Reference Standard IEC 61215 / IEC 61730